MAT2540, Classwork11, Spring2026

8.2 Solving Linear Recurrence Relations (Conti.)

17. How to guess the form of the particular solution $a_n^{(p)}$ of a linear non-homogeneous recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k} + F(n)?$$

The form of the particular solution $a_n^{(p)}$ depends on the function $F(n)$:

Non-homogeneous term $F(n)$	Trial solution for $a_n^{(p)}$
Polynomial	
$F(n) = a$ where a is a constant	$a_n^{(p)} =$ A where A is a constant
$F(n) = n$	$a_n^{(p)} =$ $An+B$ where A, B are constants
$F(n) = an + b$ where a, b are constants	$a_n^{(p)} =$ $An+B$ where A, B are constants
$F(n) = an^2 + bn + c$ where a, b, c are constants	$a_n^{(p)} =$ An^2+Bn+C where A, B, C are constants
Power/exponential functions	
$F(n) = 2^n$	$a_n^{(p)} =$ $A\cdot 2^n$ where A is a constant
$F(n) = -5^n$	$a_n^{(p)} =$ $A\cdot 5^n$ where A is a constant
$F(n) = e^n$	$a_n^{(p)} =$ $A\cdot e^n$ where A is a constant
Combination functions	
$F(n) = n + 2^n$	$a_n^{(p)} =$ $An+B+C\cdot 2^n$ where A, B, C are constants
$F(n) = (n + a)2^n$	$a_n^{(p)} =$ $(An+B)\cdot 2^n$ where $A, B,$ ~~C~~ are constants

Here, a, b, c are known constants and A, B, C are undetermined constants and can be found by using initial data

18.In general, we need to consider the characteristic roots r_0 of associated homogeneous recurrence relation:

Is $s = r_0$?	Non-homogeneous term $F(n)$	Trial solution for $a_n^{(p)}$
No	$F(n) = (b_t n^t + b_{t-1} n^{t-1} + \cdots + b_1 n^1 + b_0)s^n$	$a_n^{(p)} =$ $(P_t n^t + P_{t-1} n^{t-1} + P_{t-2} n^{t-2} + \cdots + P_2 n^2 + P_1 n + P_0)s^n$
Yes	$F(n) = (b_t n^t + b_{t-1} n^{t-1} + \cdots + b_1 n^1 + b_0)s^n$	$a_n^{(p)} =$ $n^m(P_t n^t + P_{t-1} n^{t-1} + P_{t-2} n^{t-2} + \cdots + P_2 n^2 + P_1 n + P_0)s^n$

Here m is the multiplicity of r_0.

19. Find the trial solution $a_n^{(p)}$ for $a_n = 6a_{n-1} - 9a_{n-2} + F(n)$ when (a) $F(n) = 2^n$, (b) $F(n) = (n^2 + 1)3^n$, (c) $F(n) = 3^n$, (d) $F(n) = n3^n$, (e) $F(n) = n^2 2^n$.

20. Find the solution of $a_n = 6a_{n-1} - 9a_{n-2} + 2^n$ with $a_1 = 11$ and $a_2 = 7$.

8.3 Divide-and-Conquer Algorithms and Recurrence Relations

1. Definition: Divide-and-Conquer Algorithms.

   An algorithm is called _____ algorithm if it divides a problem into one or more instances of the _____ and they conquer the problem by using the solutions of the smaller problems to find a solution of the original problem.

2. Definition: Divide-and-Conquer Recurrence Relations.

   If $f(n)$ represents the number of operations required to solve the problem of size n, it follows that f satisfies the recurrence relation

$$f(n) = \alpha f\left(\frac{n}{\beta}\right) + g(n)$$

   This is called a _____which is a recursive algorithm dividing a problem of size _____ into _____, where each subproblem is of size _____ and total of $g(n)$ extra operations which are required in the _____ of the algorithm to combine the solutions of the subproblems into a solution of the original problem.

3. Example: Binary Search.

   The problem of size n has been reduced to _____ problem of size _____ and _____ comparisons are needed:

   1) _____

   2) _____

   The divide-and-conquer recurrence relation:

$$f(n) = _____$$

ALGORITHM 3 The Binary Search Algorithm.

procedure *binary search* (x: integer, $a_1, a_2, \ldots, a_n$: increasing integers)
$i := 1$ {i is left endpoint of search interval}
$j := n$ {j is right endpoint of search interval}
while $i < j$
    $m := \lfloor (i+j)/2 \rfloor$
    if $x > a_m$ **then** $i := m + 1$
    else $j := m$
if $x = a_i$ **then** *location* $:= i$
else *location* $:= 0$
return *location* {*location* is the subscript i of the term a_i equal to x, or 0 if x is not found}

4. Example: Merge Sort.

   This algorithm splits a list to be sorted with n items into _____ lists with ____ elements each, and uses fewer than n comparisons to merge the two sorted lists of n/2 items each into one sorted list.

   The divide-and-conquer recurrence relation: $M(n) =$_____

ALGORITHM 9 A Recursive Merge Sort.

procedure *mergesort*($L = a_1, \ldots, a_n$)
if $n > 1$ **then**
    $m := \lfloor n/2 \rfloor$
    $L_1 := a_1, a_2, \ldots, a_m$
    $L_2 := a_{m+1}, a_{m+2}, \ldots, a_n$
    $L :=$ *merge*(*mergesort*(L_1), *mergesort*(L_2))
{L is now sorted into elements in nondecreasing order}

5. Given a recurrence relation of the form $f(n) = \alpha f\left(\frac{n}{\beta}\right) + g(n)$. Derive estimates of the size of functions that satisfy such recurrence relations.

   Suppose that f satisfies this recurrence relation whenever n is divisible by β. Let $n = \beta^k$, where k is a positive integer. Then